

$$\frac{\partial u}{\partial t} = \frac{\alpha}{u^2} \frac{\partial^2 u}{\partial z^2}$$

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial z} (J) \rightarrow \frac{\partial u}{\partial t} = \frac{\partial}{\partial z} \left(\frac{\alpha}{u^2} \frac{\partial u}{\partial z} \right)$$

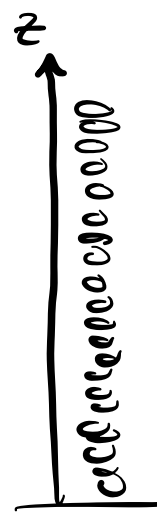
$$\frac{\partial u}{\partial t} = - \frac{2\alpha}{u^3} \frac{\partial u}{\partial z} \cdot \frac{\partial u}{\partial z} + \frac{\alpha}{u^2} \frac{\partial^2 u}{\partial z^2}$$

$$\frac{\partial u}{\partial t} = - \frac{2\alpha}{u^3} \left(\frac{\partial u}{\partial z} \right)^2 + \frac{\alpha}{u^2} \frac{\partial^2 u}{\partial z^2}$$

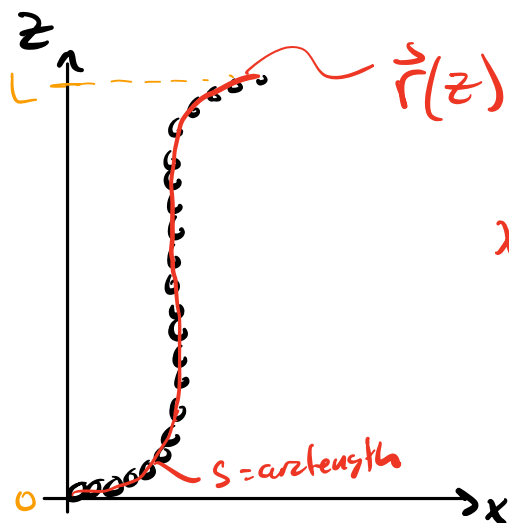
$$\frac{\partial u}{\partial z} \left\{ \begin{array}{l} \frac{\partial u_0}{\partial z} = \frac{-\frac{1}{2}u_0 + 2u_1 - \frac{1}{2}u_2}{\Delta z} \\ \frac{\partial u_1}{\partial z} = \frac{\frac{1}{2}u_2 - \frac{1}{2}u_0}{\Delta z} \\ \frac{\partial u_2}{\partial z} = \frac{\frac{1}{2}u_3 - \frac{1}{2}u_1}{\Delta z} \end{array} \right. \rightarrow D_z = \begin{bmatrix} 0 & 0 & 0 & 0 & \dots \\ \frac{1}{2} & 0 & -\frac{1}{2} & 0 & \dots \\ 0 & \frac{1}{2} & 0 & -\frac{1}{2} & \dots \end{bmatrix}$$

$$\frac{\partial u_k}{\partial z} = \frac{\frac{1}{2}u_k - \frac{1}{2}u_{k-1}}{\Delta z} = 0 \rightarrow u_k = u_{k-1}$$

$$\left\{ \begin{array}{ll} \frac{\partial u}{\partial t} = \frac{\partial}{\partial z} \left(\frac{\alpha}{u^2} \frac{\partial u}{\partial z} \right) & \text{for } z \in (0, L), t > 0 \\ \frac{\alpha}{u^2} \frac{\partial u}{\partial z} \Big|_{z=0} = 0 & \text{for } z=0, t > 0 \\ \frac{\alpha}{u^2} \frac{\partial u}{\partial z} \Big|_{z=L} = 0 & \text{for } z=L, t > 0 \\ u(z, 0) = f(z) & \text{for } t=0, z \in [0, L] \end{array} \right.$$



where $f'(0) = 0$ and $f'(L) = 0$.

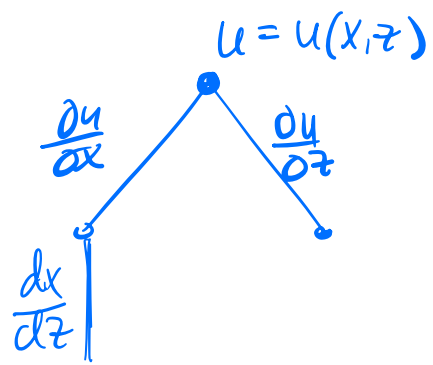


$$\vec{r}(z) = \langle x(z), z \rangle$$

$$x(z) = r_0(1 - e^{-az}) + r_L e^{a(z-L)}$$

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial s} \left(\frac{\alpha}{u^2} \frac{\partial u}{\partial s} \right)$$

$$\frac{\partial u}{\partial t} =$$



$$\frac{\partial u}{\partial x} \cdot \frac{dx}{dz} + \frac{\partial u}{\partial z}$$